

Trainings ▾

General Information

The purpose of the variable geometry turbocharger (VGT):

- When the exhaust gas recirculation (EGR) flow is commanded, the VGT will close the nozzle in the turbine housing to create more back pressure in the exhaust manifold, to force the EGR flow back into the engine.
- Closing the VGT will also increase exhaust gas temperature under certain normal engine operating conditions and during aftertreatment regeneration events.
- The turbocharger also functions to improve engine performance by building boost more quickly during acceleration.

Use the following procedure for further information regarding VGT and aftertreatment system interactions. Refer to Procedure 011-999 in Section F. (</qs3/pubsys2/xml/en/procedures/10/10-011-999.html>)

This procedure provides service information for the electronically controlled VGT actuators.

Note : To avoid improper installation of the turbocharger actuator, follow each step in its entirety and in the proper sequence.

The starting motor wiring can electrically interfere with the VGT pigtail and wiring harness. The starting motor wiring **must** be at least 152 mm [6 in] away from the actuator and harness.

Preparatory Steps

WARNING

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

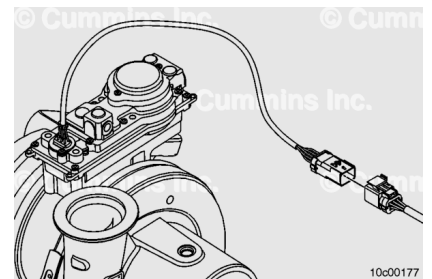
WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Clean the area around the turbocharger actuator and dry with compressed air.
- Drain the engine coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Disconnect the turbocharger actuator coolant supply and return lines from the turbocharger actuator. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/10/10-010-041-tr.html)
- Remove the turbocharger actuator wiring harness zip ties.

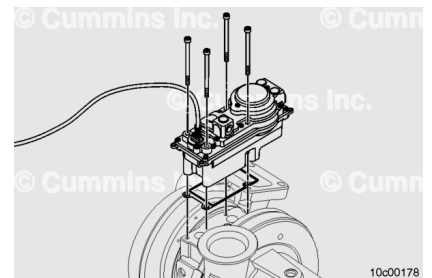
Remove

Disconnect the wiring harness from the turbocharger actuator by sliding the locking tang to the open position, and then pushing down on the release lever and pulling the connection apart.



⚠ CAUTION ⚠

Make sure the coolant has been drained before the actuator is removed. Failure to do so can result in damage to the actuator.



Remove the four turbocharger actuator mounting capscrews with a 5-mm Allen wrench and discard the capscrews.

Remove the actuator.

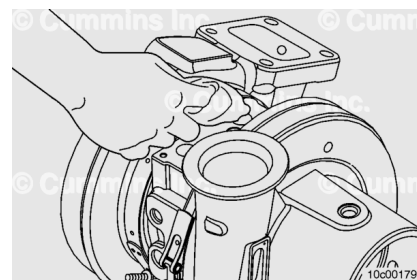
Remove and discard the turbocharger actuator sealing gasket.

Clean and Inspect for Reuse

Use a clean cloth to wipe around the surface where the turbocharger actuator seals to the turbocharger.

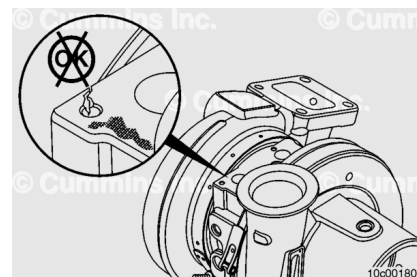
Note : Take care **not** to drop any dirt or debris into the turbocharger.

Inspect the sealing surfaces and the sealing gasket for signs of leaks, cracks, or other damage.



If the sealing surface of the turbocharger is damaged and causing a leak, the turbocharger **must** be replaced.
Discontinue work with this procedure. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)

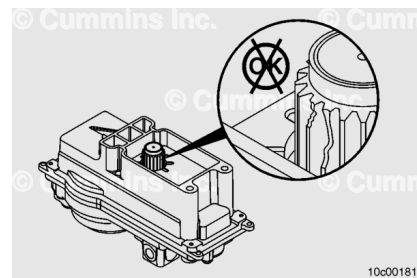
If the sealing surface of the actuator is causing a leak, replace **only** the actuator.



Do **not** turn the gear.

Inspect the pinion gear on the actuator for excessive wear or damaged teeth.

Replace the actuator if the teeth are worn or damaged.

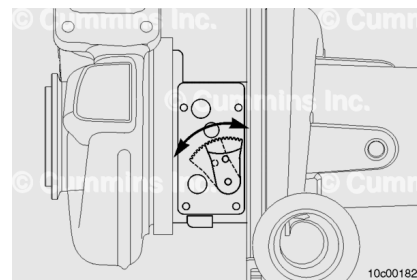


Inspect the sector gear on the turbocharger for excessive wear, damaged teeth, or a broken shaft.

Grasp the sector gear by hand and move it through its operational range. Do **not** use any tools.

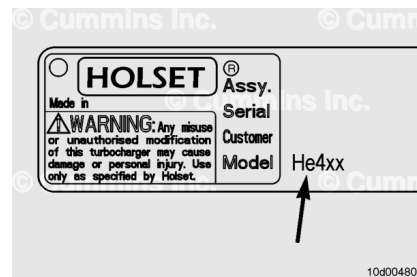
Within the turbocharger, there are solid end stops which limit sector gear travel in both directions.

The sector gear **must** move by hand through the entire range of motion.

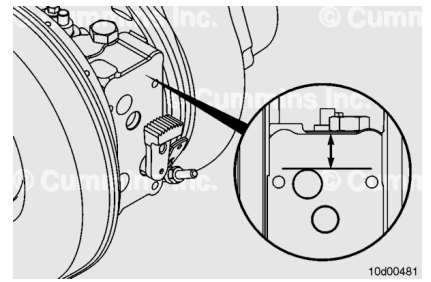


The sector gear travel gauges are designed for specific turbocharger models. To determine the correct gauge, verify the turbocharger model number from the turbocharger data plate. Select the gauge that matches the first three letters of the turbocharger model number.

Example: HE5xx, HE4xx, and so on.

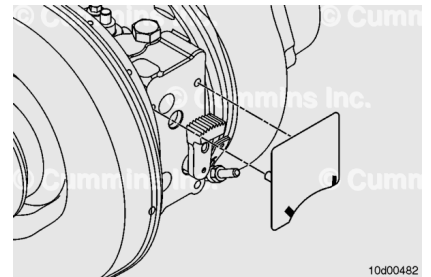


An alternate method of determining the turbocharger model number, if the turbocharger data tag is missing or covered with paint, is to measure the distance between the top of the cup plug hole to the bearing housing edge. If the distance is greater than 19 mm [0.748 in], the turbocharger model is HE4xx. If the distance is less than 19 mm [0.748 in], the model is a HE5xx.

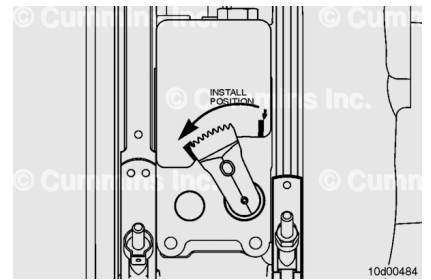


To install the sector gear travel gauge, carefully bend the gauge and slide the thin section under the sector gear. If necessary, pull the sector gear out by hand to allow more clearance for the gauge.

Verify that the three alignment bosses are fully engaged in the bearing housing.



Rotate the sector gear **counterclockwise** toward the turbocharger turbine housing until it stops. The edge of the sector gear **must** be in the green acceptance zone of the gauge.

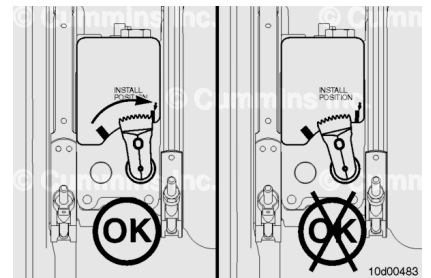


Rotate the sector gear **clockwise** toward the turbocharger compressor housing until it stops. The edge of the sector gear **must** be in the green acceptance zone of the gauge.

If the sector gear does **not** go through the full range of motion, or if the sector gear requires excessive force to move it by hand, do the following:

Use the following procedure in the Signature™, ISX, and QSX15 Service Manual, Bulletin 3666239. Refer to Procedure 010-145 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-145.html>)

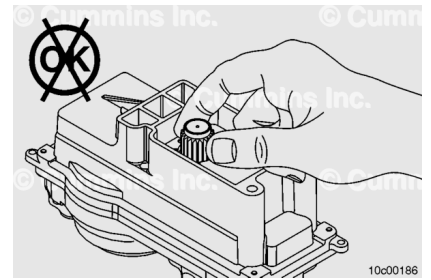
Use the following procedure in the ISM, ISMe, and QSM11 Service Manual, Bulletin 3666322. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html>)



Install

⚠ CAUTION ⚠

During the turbocharger actuator install procedure, the pinion gear on the actuator will move. Keep hands and tools away from the pinion gear during the procedure.

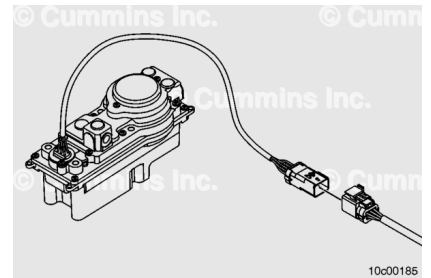


Following these instructions in order is very important.

- Continue through the entire turbocharger actuator installation procedure before attempting to troubleshoot any other fault codes.
- Verify that the turbocharger actuator is removed from the turbocharger bearing housing.
- Verify that the turbocharger actuator electrical connector is disconnected from the engine wiring harness.
- Turn the keyswitch ON. Connect INSITE™ electronic service tool and wait 60 seconds.

Note : It is important that the keyswitch remain in the ON position throughout both the Install and Calibrate procedures unless otherwise directed.

- Connect the turbocharger actuator electrical connector to the engine wiring harness.



If Fault Code 2634 becomes active, disconnect the turbocharger actuator connector from the engine wiring harness with the keyswitch ON. Connect the turbocharger actuator electrical connector. Fault Code 2634 will go inactive.

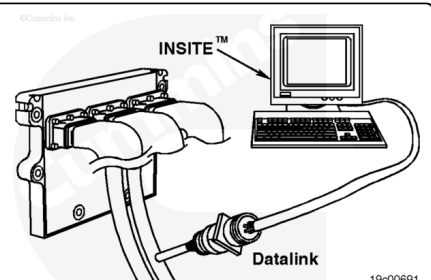
It is normal and expected to have Fault Code 2449 active when a new turbocharger actuator is connected to the engine, because it is **not** calibrated to the turbocharger.

Continue through the engine turbocharger actuator installation procedure before attempting to troubleshoot any other fault codes.

Use INSITE™ electronic service tool and go to the ECM Diagnostic Tests screen.

From the list, select VGT Electronic Actuator Installation and Calibration and click on the “next” button.

Note : The VGT Electronic Actuator Installation and Calibration is not a diagnostic test. It is the procedure



to properly install and calibrate the turbocharger actuator. Running this procedure improperly can result in additional fault codes and/or damage to the turbocharger or actuator.

INSITE™ electronic service tool “INSTALL ACTUATOR” command **must only** be run with the actuator **not** mounted to the turbocharger.

Locate the column labeled “Value” and left click on the down arrow. Select “INSTALL ACTUATOR” and select “START”.

This will set the actuator pinion gear to a known position to prepare it for installation to the turbocharger. This step should take less than 30 seconds to complete with INSITE™ electronic service tool.

INSITE™ electronic service tool will indicate when this step is complete.

Fault Code 2449 will be active at this point in the procedure. Continue through the turbocharger actuator procedure before troubleshooting fault codes.

If, at any point, INSITE™ electronic service tool status message indicates the procedure was stopped or failed, leave the keyswitch ON and cycle the power to the actuator by unplugging it from the harness, and then reconnecting it.

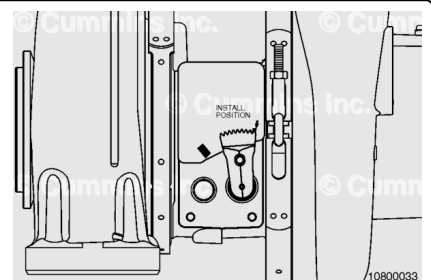
If cycling the power to the actuator does **not** work, unplug the actuator, turn the keyswitch OFF for 30 seconds, and then restart INSITE™ electronic service tool. Then start over, beginning with the “INSTALL ACTUATOR” step.

Repeat process three times. If still **not** working replace actuator.

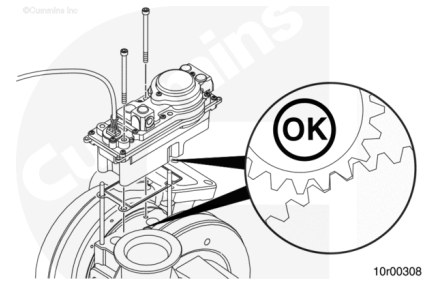
Grasp the sector gear by hand and rotate the sector gear toward the turbocharger compressor housing. Make sure the edge of the sector gear is rotated all the way toward the “Install Position” arrow.

Coat the teeth on the sector gear with the grease packet supplied in the installation kit.

Note : It is critical for smooth reliable operation of the actuator to use the full amount of the Holset® supplied grease.



Do not attempt to force the actuator onto the bearing housing by using the capscrews. Misalignment can cause damage to the actuator or turbocharger.



Align the actuator and gasket assembly with the turbocharger bearing housing face. Insert the threaded guide pins into the diagonally opposite fastener holes in the bearing housing and locate the turbocharger actuator onto the guide pins.

Slide the turbocharger actuator into position, verify the pilot bearing is aligned with the mating hole in the bearing housing and the output gear of the actuator is smoothly engaged with the turbocharger sector gear.

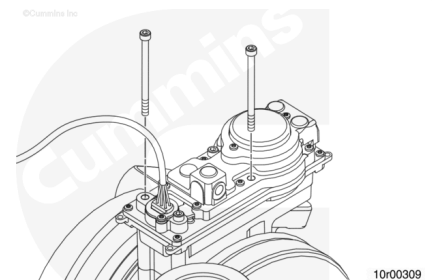
Insert the actuator socket head capscrews in the remaining fastener holes and hand-tighten.

The actuator pinion gear and the turbocharger sector gear **must** engage smoothly.

With the actuator aligned on the bearing housing, replace the guide pins with the turbocharger actuator capscrews.

Install the remaining two capscrews.

Tighten the four capscrews in a criss-cross pattern.



Torque Value:

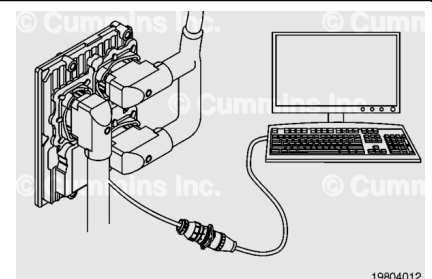
- 1. 3 n•m [27 in-lb]
- 2. 11 n•m [97 in-lb]

Calibrate

The turbocharger actuator **must** be calibrated to the turbocharger. This step **must** be performed to make sure proper turbocharger operation is achieved.

Calibration instructions:

Within the INSITE™ electronic service tool screen labeled “VGT Electric Actuator Install and Calibrate”, locate the column labeled “Value” and left click on the down arrow. Select “CALIBRATE ACTUATOR”.



INSITE™ electronic service tool “CALIBRATE ACTUATOR” command **must only** be run with the actuator mounted to the turbocharger.

Follow the instruction on the screen in order to calibrate the turbocharger actuator to the turbocharger. INSITE™ electronic service tool will indicate when this step is complete.

If INSITE™ electronic service tool status message indicates the procedure was stopped or failed, turn the keyswitch OFF for 30 seconds and key back ON. Then start over, beginning with the “INSTALL ACTUATOR”

It is normal to have an active Fault Code 2387 at this point.

Turn the keyswitch in the OFF position for 30 seconds. Then key ON and refresh the fault code screen. All turbocharger actuator fault codes should be inactive. Use INSITE™ electronic service tool to clear all fault codes.

Note : Make sure all fault codes are inactive and cleared before adding coolant to the engine.

Finishing Steps

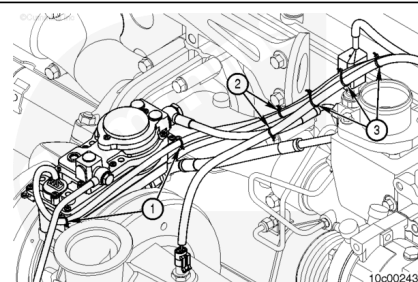
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the turbocharger actuator coolant supply and return lines. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-041-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Operate the engine to make sure the turbocharger actuator operates properly and all turbocharger actuator fault codes are inactive.
- Check the engine for leaks.

The starting motor wiring can electrically interfere with the VGT pigtail and wiring harness. The starting motor wiring **must** be at least 152 mm [6 in] away from the actuator and harness.

- Secure the turbocharger actuator wiring away from hot and abrasive parts.



- Zip tie the turbocharger actuator harness to the two ears on the actuator housing (1).
- Secure the turbocharger actuator harness, turbocharger compressor inlet temperature sensor harness, and the turbocharger speed sensor harness to the turbocharger coolant outlet hose (2). In order to prevent the wire harness from being damaged, use two zip ties as shown.
- Secure the turbocharger actuator, speed sensor, and compressor inlet temperature sensor wiring harness as indicated (3).